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# Thermally Tunable Photonic Synaptic Transistor with Bioinspired Temporal Dynamics for Task-Adaptive Neuromorphic Vision

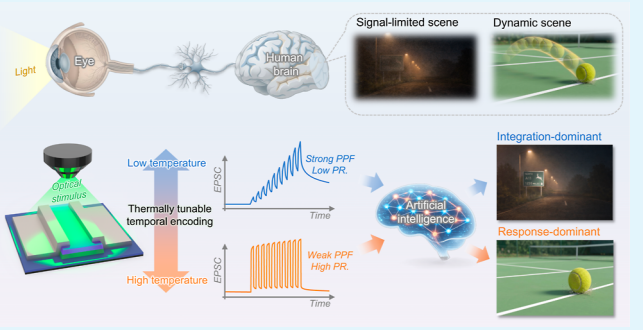
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**ABSTRACT:** Inspired by temperature-associated temporal redistribution in biological systems, we demonstrate a thermally tunable photonic synaptic transistor (TT-PST) based on a magnesium-doped zinc oxide (MZO)/indium−gallium−zinc oxide (IGZO) heterostructure. By using temperature as a simple external control parameter, we enable continuous and reversible reshaping of synaptic temporal behavior without structural reconfiguration. As the temperature increases from 20 to 80 °C, the maximum photoresponsivity (Max. PR) rises from 0.43 to 2.25 A/W, while the temporal persistence decreases sharply, resulting in a transition from integration-dominant to response-dominant operation. Leveraging this capability, temperature-dependent temporal encoding is evaluated in two representative visual tasks using experimentally extracted device dynamics in a convolutional neural network (CNN) framework. Weak-signal recognition under low-light conditions achieves higher accuracy at 20 °C (82.86%) compared to 80 °C (69.23%), whereas dynamic motion analysis shows superior performance at 80 °C (88.03%) versus 20 °C (74.37%). These findings establish temperature as a bioinspired, physically accessible knob for on-demand temporal dynamics engineering in a single hardware platform, providing an efficient strategy for task-adaptive neuromorphic vision systems.




**KEYWORDS:** *photonic synaptic transistor, neuromorphic vision, temporal dynamics engineering, temperature-dependent modulation, bioinspired temporal encoding*

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## 1. INTRODUCTION

Photonic synaptic devices have attracted considerable attention in neuromorphic vision due to their capability to integrate sensing and temporal information processing within a single device.[1,2] In practical visual tasks, however, different scenarios require distinct temporal characteristics.[3,4] Integration-demanding conditions, such as low-visibility or signal-limited environments, benefit from prolonged temporal accumulation, whereas response-demanding conditions, such as motion-rich or rapidly changing scenes, require fast response with minimal temporal persistence.[5,6] Rather than representing a strict binary distinction, these conditions correspond to different points along a continuum of temporal demand. Prior studies have demonstrated photonic synaptic devices optimized for specific temporal regimes, such as strong retention or fast response.[7−10] Recent studies have also explored gate-tunable persistent photocurrent dynamics in neuromorphic optoelectronic devices, where thermal effects influence the temporal photoresponse behavior.[11] Nevertheless, approaches that allow continuous and reversible tuning of temporal dynamics within a single platform remain under active exploration.

In biological systems, temporal dynamics are often modulated under environmental and physiological conditions. As illustrated in Figure 1a, hawkmoths provide an example of such adaptability. During low-activity states, their sensory system employs longer temporal integration to enhance weak-signal perception, corresponding to an integration-dominant mode. In contrast, under threat or high-activity conditions, hawkmoths elevate their body temperature through rapid wing vibrations, leading to faster sensory responses and reduced temporal persistence, corresponding to a response-dominant mode. This thermoregulatory behavior is associated with increased activity levels and higher information transmission rates in visual neurons, which support predator avoidance. Importantly, this temperature dependence suggests a continuous redistribution of temporal processing between sensitivity and temporal resolution rather than indicating a discrete switch between predefined modes.[12,13] These observations highlight that external factors, including temperature, can serve as effective modulators for

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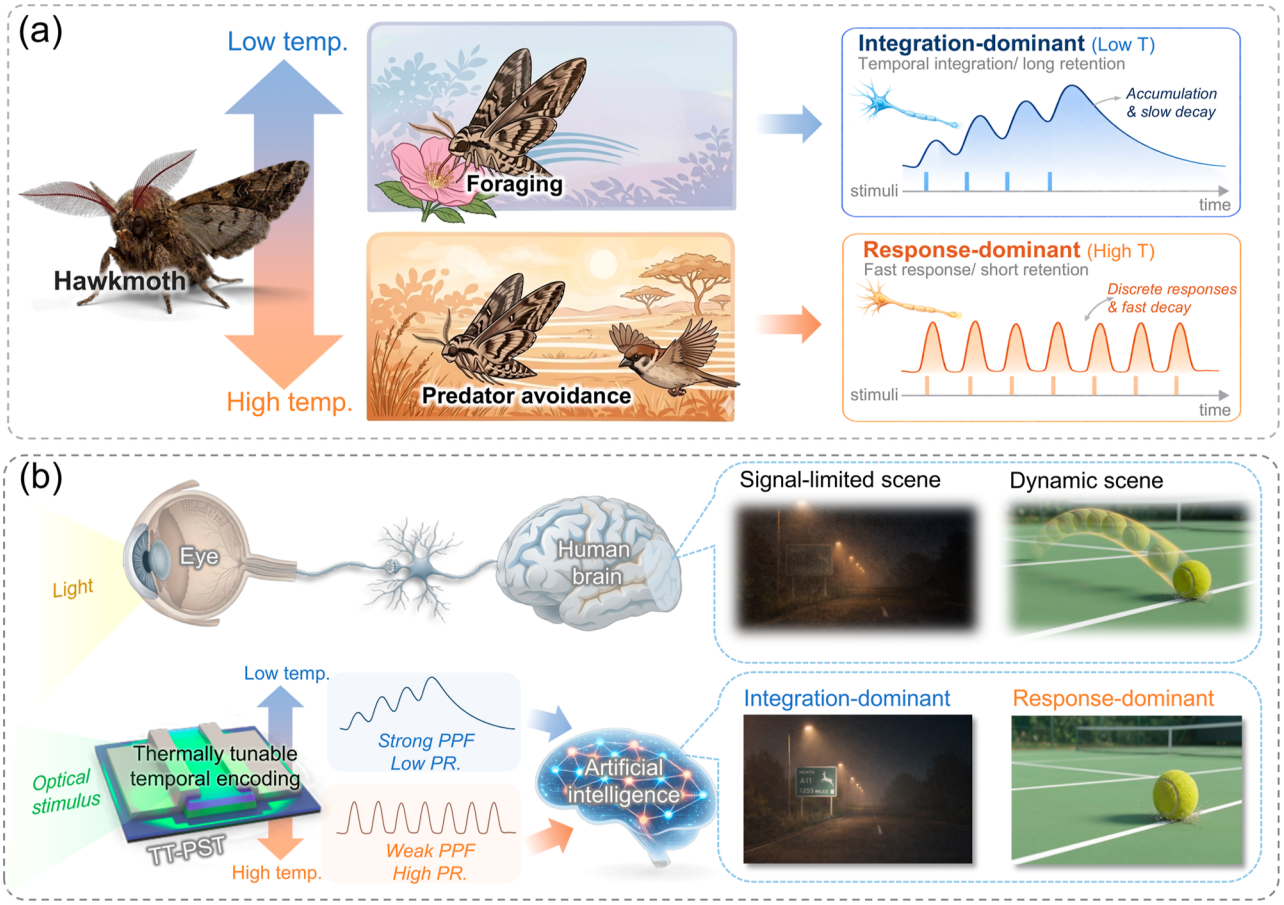



**Figure 1.** (a) Illustration of hawkmoths behavioral states under different temperature conditions and their corresponding temporal response modes. (b) Comparison between human vision and TT-PST-based perception in different scenarios.

dynamically reshaping the temporal characteristics in biological vision.[14,15]

Motivated by this bioinspired temperature-dependent temporal modulation, we propose a thermally tunable photonic synaptic transistor (TT-PST) based on a magnesium-doped zinc oxide (MZO)/indium−gallium−zinc oxide (IGZO) heterostructure.[16−21] By introducing temperature as an external control parameter, the device enables continuous tuning of its temporal response within a single platform so that it can transition between temporal integration and fast response. To characterize this temperature-dependent behavior, we investigate the temporal dynamics through optoelectronic synaptic measurements, including excitatory postsynaptic current (EPSC), paired-pulse facilitation (PPF), and photoresponsivity (PR). Two representative visual scenarios—signal-limited perception and dynamic visual processing—are employed to explore the correspondence between temporal regimes and task requirements, as illustrated in Figure 1b. Array-level operation and neural-network-based recognition are further investigated to evaluate the potential for task-adaptive neuromorphic vision.[22−26]

## 2. RESULTS AND DISCUSSION

### 2.1. Device Architecture and Material Characterization

The proposed TT-PST adopts a vertically stacked architecture, as illustrated in Figure 2a. The device consists of a heavily doped silicon substrate with a thermally grown $SiO_2$ dielectric layer ($P^+Si/SiO_2$), an amorphous IGZO channel layer, an overlying MZO functional layer, and aluminum (Al) source/drain electrodes. Figure 2b shows a top-view optical micrograph with uniform device geometry. The channel length and width are approximately 150 and 1000 $\mu$m, respectively. The cross-sectional scanning electron microscopy (SEM) image (Figure 2c) further shows the continuous stacking of the MZO and IGZO layers with a clear interface. The fabrication process is summarized in Figure S1, with detailed procedures provided in the Experimental Methods section. Photoluminescence (PL) spectra (Figure 2d) were employed to characterize the ZnO and MgZnO films with varying Mg contents. All samples exhibit two characteristic emission features: a near-band-edge (NBE) emission at approximately 390 nm and a broad deep-level emission (DLE) band ranging from 500 to 600 nm. Notably, while the NBE peak position remains nearly constant with increasing Mg content, the DLE band undergoes significant intensification and broadening so that it reaches its maximum at 10% Mg.[27] This evolution indicates that the introduction of Mg effectively increases the density and energy distribution of defect-related localized states (e.g., oxygen vacancies) within the bandgap, thereby facilitating carrier trapping processes.[28] A further increase in Mg content to 15% leads to a decline in DLE intensity because excessive doping induces nonradiative recombination centers.[29] Consequently, the 10% Mg composition was selected as the MZO layer for subsequent device investigations. The optical properties of IGZO and MZO films were characterized using ultraviolet−visible (UV−vis) absorption spectroscopy (Figure 2e). Both materials exhibit a clear


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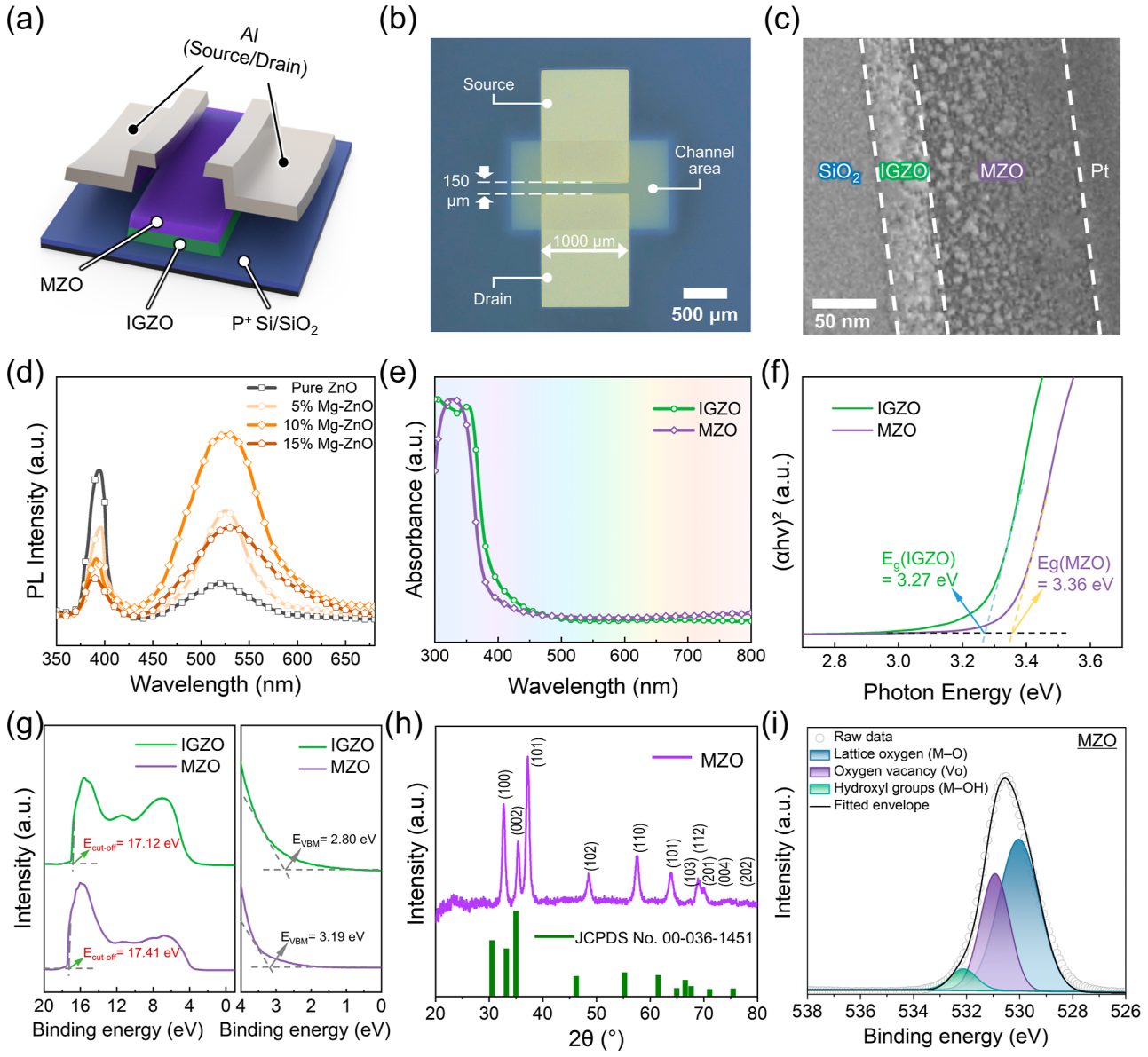



**Figure 2.** Structural architecture and material characterization. (a) Schematic illustration of the vertically stacked TT-PST. (b) Top-view optical microscope image of the device. (c) Cross-sectional SEM image of the TT-PST. (d) PL spectra of ZnO and MgZnO films with different Mg contents. (e) UV−vis absorption spectra of IGZO and MZO films. (f) Tauc plots used to extract the optical bandgaps of IGZO and MZO. (g) UPS spectra showing the secondary electron cutoff and valence band onset of IGZO and MZO. (h) XRD pattern of the MZO film. (i) Deconvoluted O 1s XPS spectrum of the MZO layer.

absorption edge in the ultraviolet region, while the normalized absorption in the visible range remains weak because they exhibit a wide-bandgap semiconductor behavior. The corresponding Tauc plots (Figure 2f) yield optical bandgaps of approximately 3.27 eV for IGZO and 3.36 eV for MZO. The electronic energy levels of IGZO and MZO were determined by ultraviolet photoelectron spectroscopy (UPS), as shown in Figure 2g. The secondary electron cutoff energies ($E_{cutoff}$) are approximately 17.12 eV for IGZO and 17.41 eV for MZO, corresponding to work functions of about 4.10 and 3.81 eV, respectively, based on $\Phi = h\nu - E_{cutoff}$ with $h\nu = 21.22$ eV. From the valence band onset, the valence band maxima ($E_{VBM}$) relative to the Fermi level are estimated to be 2.80 eV for IGZO and 3.19 eV for MZO. Combined with the optical bandgaps obtained from the Tauc plots, these values define the band structures of the two materials and the corresponding interfacial band alignment, as summarized in Figure S2. The XRD pattern of MZO (Figure 2h) is consistent with the wurtzite structure, with the main diffraction peaks indexed to the (100), (002), and (101) planes (JCPDS No. 00-036-1451).[30−32] No additional peaks associated with secondary phases are observed, which indicates that Mg incorporation does not alter the primary crystal phase. The slight broadening of the (002) peak suggests reduced crystallite size or increased lattice disorder. The oxygen-related chemical states of MZO were analyzed by deconvolution of the high-resolution O 1s spectrum (Figure 2i). The spectrum is fitted with three components centered at approximately 530.0, 531.0, and 532.0 eV, corresponding to lattice oxygen (M−O), oxygen-vacancy-related states ($V_O$), and surface hydroxyl species (M−OH), respectively. The relative fractions of these components are 60.9, 33.8, and 5.3%, respectively, with the $V_O$-related component constituting a significant portion of the total


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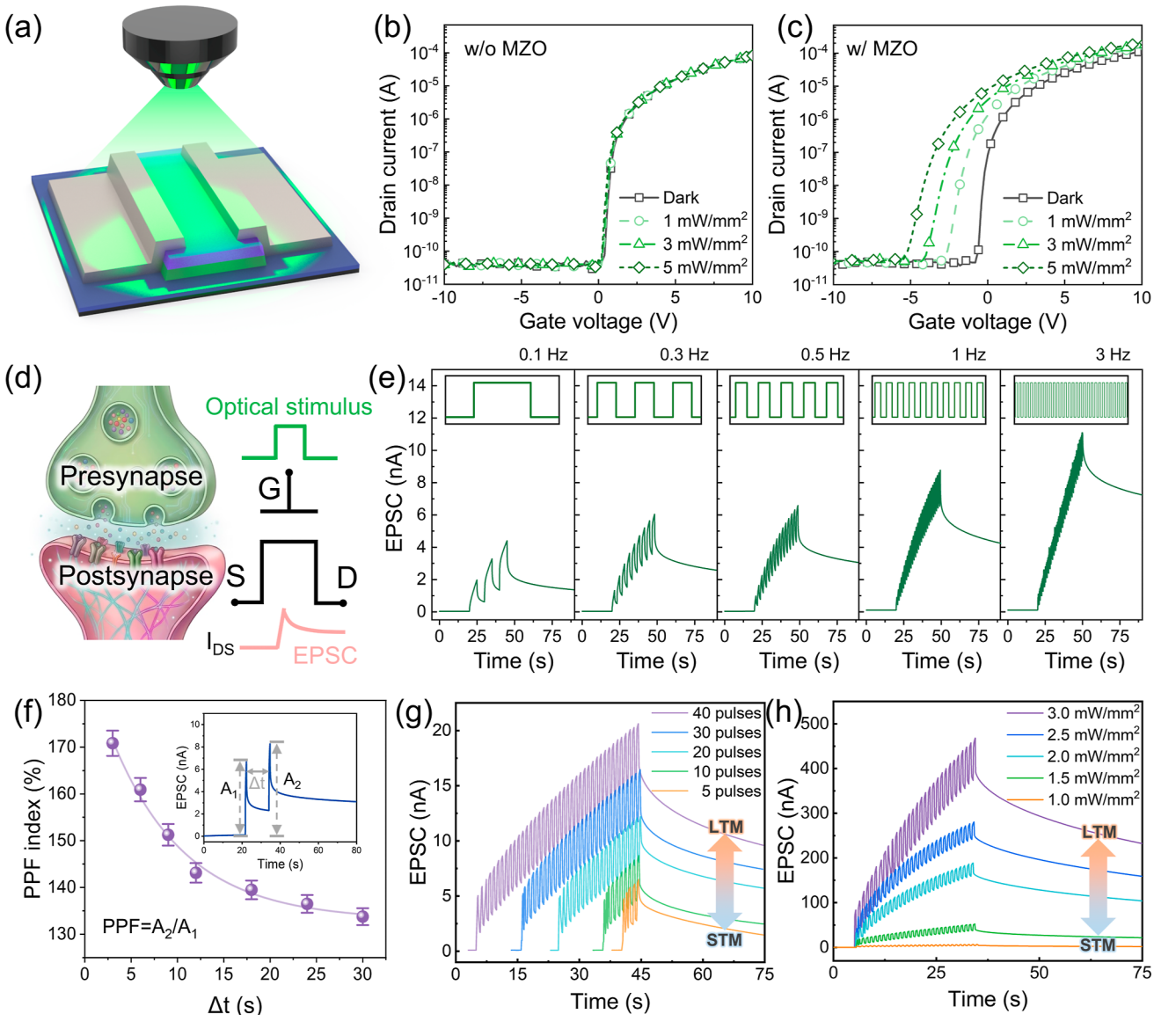



**Figure 3.** Synaptic response and temporal integration behavior of the TT-PST. (a) Schematic illustration of the device operation under green-light illumination. (b) Transfer characteristics of the reference device without the MZO layer under different light intensities. (c) Transfer characteristics of the TT-PST with the MZO layer under different light intensities. (d) Schematic illustration of the optoelectronic synaptic operation. (e) EPSC responses under pulsed optical stimulation at different frequencies (0.1−3 Hz). (f) PPF index as a function of pulse interval $\Delta t$. Error bars are derived from nine devices. (g) EPSC responses under different numbers of optical pulses (5−40 pulses) at a fixed light intensity. (h) EPSC responses under different light intensities (1.0−3.0 mW/mm$^2$) at fixed pulse conditions.

spectral weight.[33] This distribution indicates that $V_O$-related states form a major part of the defect landscape in MZO and provide abundant sites for carrier trapping.[34]

## 2.2. Synaptic Response and Temporal Integration Behavior

As shown in Figure 3a, the TT-PST operates under green-light illumination (532 nm), which falls within the absorption range of the device materials. For comparison, a reference device without the MZO layer was also measured. The reference device (Figure 3b) was measured with a drain bias ($V_{DS}$) of 10.1 V and a gate voltage ($V_G$) swept from −10 to 10 V, where the relatively higher $V_{DS}$ placed the device in the saturation regime, enabling clearer observation of the transfer modulation behavior under optical illumination while maintaining sufficient readout current. The device exhibited minimal modulation under varying light intensities, with the transfer curves largely overlapping, which indicates a weak response under green-light illumination. In contrast, the TT-PST (Figure 3c) shows a pronounced negative shift in threshold voltage under illumination, accompanied by a substantial increase in drain current with increasing light intensity. The transfer curves display a clear and systematic separation as the optical input increases. This comparison demonstrates that the incorporation of the MZO layer fundamentally alters the modulation behavior under visible illumination, enabling a transition from weak perturbation to a controllable current modulation process. Figure 3d illustrates the operating analogy between the TT-PST and a biological synapse, where the optical pulse acts as the presynaptic stimulus and the transient drain current response is treated as the EPSC. Under pulsed optical stimulation, the TT-PST exhibits characteristic synaptic responses, with each light pulse inducing a distinct EPSC signal. Figure 3e shows the EPSC responses measured at a fixed light intensity of 1 mW/mm$^2$, with the device


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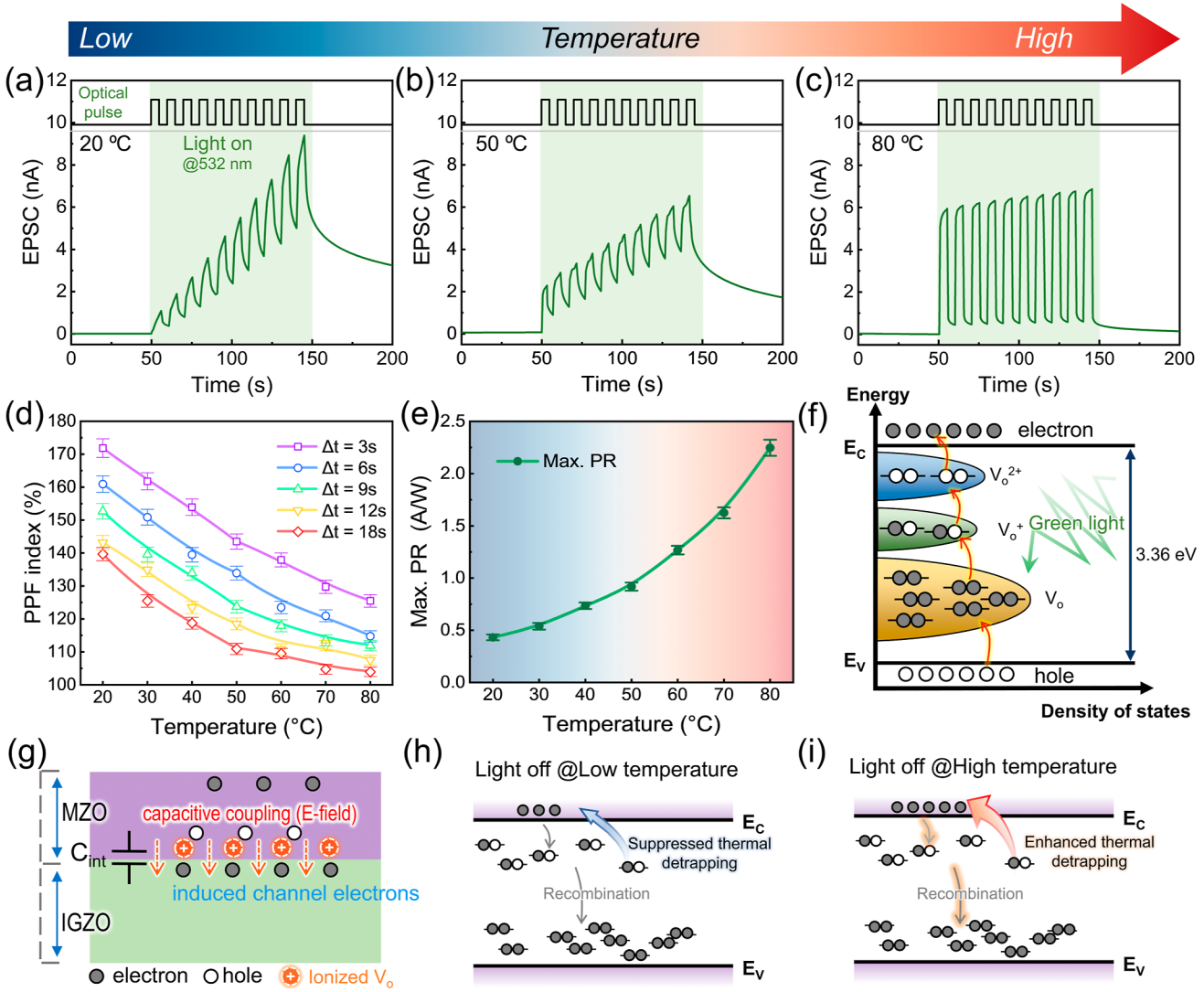



**Figure 4.** Temperature-dependent synaptic dynamics and mechanism. (a–c) EPSC responses under pulsed green-light illumination (1 mW/mm$^2$) at different temperatures (20, 50, and 80 °C). (d) PPF index as a function of temperature for different pulse intervals ($\Delta t$). Error bars are derived from nine devices. (e) Max. PR as a function of temperature. Error bars are derived from nine devices. (f) Schematic energy diagram illustrating subgap-state-assisted photoexcitation in the MZO layer. (g) Schematic illustration of interfacial electric field formation via ionized defect states and capacitive coupling to the IGZO channel. (h,i) Schematic illustrations of carrier relaxation after light-off at low and high temperatures, showing suppressed and enhanced thermal detrapping, respectively.

biased at $V_G$ = −2 V and $V_{DS}$ = 1.5 V, where the relatively low $V_{DS}$ keeps the device operating in the linear regime for stable low-power synaptic measurements throughout the transient response experiments. At a low stimulation frequency of 0.1 Hz, the responses to individual pulses remain largely separated. As the frequency increases from 0.3 to 3 Hz, the EPSC progressively accumulates, giving rise to a pronounced temporal integration behavior. To quantify the correlation between two successive pulses, the PPF index is expressed as PPF = $A_2/A_1$, where $A_1$ and $A_2$ are the peak amplitudes of the first and second EPSC responses, respectively. As shown in Figure 3f, the PPF index decreases monotonically with increasing pulse interval $\Delta t$, with the averaged response from nine devices exhibiting a consistent reduction from 170.3% at $\Delta t$ = 3 s to 134.1% at $\Delta t$ = 30 s and only limited device-to-device variation throughout the measured range. This gradual reduction reflects the weakening temporal correlation between successive stimuli as the interval increases. The temporal response of the TT-PST was further examined by varying both the number of optical pulses and the input light intensity. As shown in Figure 3g, the measurements were conducted at a fixed light intensity of 1 mW/mm$^2$, while maintaining $V_G$ = −2 V and $V_{DS}$ = 1.5 V. Increasing the number of pulses from 5 to 40 leads to a progressive increase in the EPSC amplitude, followed by a gradual decay after the termination of stimulation. The accumulated response becomes more pronounced with increasing pulse number, indicating a transition from short-term memory (STM)-like to long-term memory (LTM)-like behavior. Figure 3h shows that, under identical pulse conditions and the same bias configuration, increasing the light intensity (1.0–3.0 mW/mm$^2$) results in a substantial enhancement of the EPSC amplitude, while the overall decay profile remains comparable. A similar STM-to-LTM-like evolution can be observed with increasing optical input, suggesting that both pulse number and light intensity can modulate the effective memory depth of the device. These results demonstrate that the TT-PST exhibits a tunable temporal response under optical stimulation, with its memory


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characteristics continuously adjustable through both temporal and intensity domains.

### 2.3. Temperature-Modulated Synaptic Dynamics and Kinetic Mechanism

The temperature-dependent synaptic response of the TT-PST was investigated under pulsed green-light illumination (532 nm, 1 mW/mm$^2$), with the device biased at $V_G = -2$ V and $V_{DS} = 1.5$ V, as shown in Figure 4a−c. At 20 °C, the EPSC exhibits a gradual accumulation over successive pulses within the illumination window (50−150 s), forming a progressive increase in amplitude and decaying slowly after the light is turned off, indicating pronounced temporal retention. As the temperature increases to 50 °C, both the accumulation and retention are reduced, accompanied by a faster decay process. At 80 °C, the response transitions to nearly isolated transient pulses with minimal interpulse accumulation, and the postillumination current rapidly returns to the baseline. This progressive evolution reflects a systematic shortening of the characteristic response time scale with increasing temperature, leading to a transition from temporally integrated behavior to a fast-response regime. The temperature-dependent modulation was further quantified through the PPF index and the PR. As shown in Figure 4d, the PPF index decreases monotonically with increasing temperature across all pulse intervals, indicating a reduced temporal correlation between successive stimuli and a shortened effective retention time. A similar temperature-dependent evolution with a relatively small device-to-device variation was consistently observed across the nine measured devices. The PR is defined as

$$\mathrm{PR} = \frac{I_{\mathrm{light}} - I_{\mathrm{dark}}}{P_{\mathrm{in}} \cdot A_{\mathrm{p}}} \tag{1}$$

Here, $I_{\mathrm{light}}$ and $I_{\mathrm{dark}}$ denote the peak drain current measured during and prior to a single optical pulse, respectively, $P_{\mathrm{in}}$ is the incident light power density, and $A_{\mathrm{p}}$ is the effective photosensitive area.[35] As shown in Figure 4e, the maximum photoresponsivity (Max. PR) increases steadily with temperature, with the averaged values from nine devices rising from 0.43 A/W at 20 °C to 2.25 A/W at 80 °C while maintaining a relatively narrow distribution throughout the measured temperature range. The opposite trends of PPF and PR reveal a redistribution of the synaptic response characteristics, where temporal retention is progressively suppressed while instantaneous response is enhanced. As illustrated in Figure 4f, under green-light illumination, photoexcitation is dominated by subgap transitions associated with defect states in the MZO layer. Electrons are excited from localized states near the valence band toward oxygen-vacancy-related levels, accompanied by the ionization of oxygen vacancies, resulting in positively charged defect centers and a nonequilibrium carrier distribution within the MZO layer. Wavelength-dependent transfer characteristics were additionally measured under blue-, green-, and red-light illumination (405, 532, and 635 nm). As shown in Figure S3, the photoresponse gradually decreases as the illumination wavelength increases, while clear transfer modulation remains observable across the visible range. Considering the wide optical bandgaps of IGZO and MZO and their weak visible-range absorption, the observed wavelength-dependent response is consistent with defect-assisted subgap excitation associated with oxygen-vacancy-related states in the MZO layer. As shown in Figure 4g, ionized oxygen vacancies in the MZO layer form spatially distributed positive charge centers near the interface, giving rise to an interfacial electric field that modulates the channel conductance via capacitive coupling. The persistence of this modulation is governed by the occupancy of defect states. After the optical pulses are turned off, the current decay is controlled by thermally activated detrapping of electrons from oxygen-vacancy-related states (Figure 4h,i). Trapped electrons overcome an energy barrier, return to the conduction band, and recombine with positively charged vacancies, leading to the gradual neutralization of local positive charge and a corresponding weakening of the interfacial electric field, with the channel conductance recovering toward its initial state. Temperature determines the rate of this detrapping process and thus the effective lifetime of the defect charge state. At low temperature, suppressed detrapping sustains the interfacial field and promotes current accumulation, whereas at elevated temperature, accelerated detrapping shortens the lifetime of the field modulation, resulting in faster response and reduced accumulation.[36] As shown in Figure S4a, the retention time extracted from the EPSC decay behavior, defined as the time required for the EPSC to decay to 1/e of its peak value after light-off, decreases systematically with increasing temperature. The extracted values were averaged from nine devices, with relatively small device-to-device variation observed across the measured temperature range. The corresponding Arrhenius-like plot in Figure S4b further reveals an approximately linear temperature dependence, supporting a thermally activated relaxation process consistent with the proposed detrapping of electrons from oxygen-vacancy-related states. Other temperature-dependent effects may also participate in the oxide device response: channel transport and contact effects can influence the absolute current level, while adsorption/desorption of oxygen- or hydroxyl-related surface species in ZnO-based oxides can modulate the surface charge environment and photoresponse amplitude. These effects mainly affect current magnitude or surface charge balance, whereas the systematic shortening of the EPSC decay persistence with temperature is more closely associated with thermally activated relaxation dynamics. To track the reversibility of the temperature-dependent synaptic modulation along the thermal path, same-device thermal-cycling measurements were sequentially performed following the sequence of 20 → 50 → 80 → 50 → 20 °C after thermal stabilization at each temperature step. The thermal-cycling measurement was repeated 9 times on the same device, and the error bars represent the corresponding standard deviation. As summarized in Figure S5, the extracted Max. PR, PPF index, retention time, and peak EPSC exhibit closely matched values between the forward and return thermal paths. Here, the peak EPSC and retention time were extracted from the single-pulse response, with the retention time defined as the time required for the EPSC to decay to 1/e of its peak value after light-off. In particular, the returned 50 and 20 °C states exhibit values close to those measured at the corresponding temperatures during the heating process, indicating largely reversible modulation with limited hysteresis and drift under the present measurement conditions. The relatively small variations further indicate good cycle-to-cycle reproducibility and thermal stability of the temperature-dependent synaptic characteristics.

### 2.4. Array-Level Demonstration and Temperature-Dependent Memory Behavior

Based on the proposed TT-PST device, a 6 × 6 array was fabricated for array-level demonstration. As illustrated in Figure 5a, the array consists of a Mo bottom electrode, an $HfO_x$


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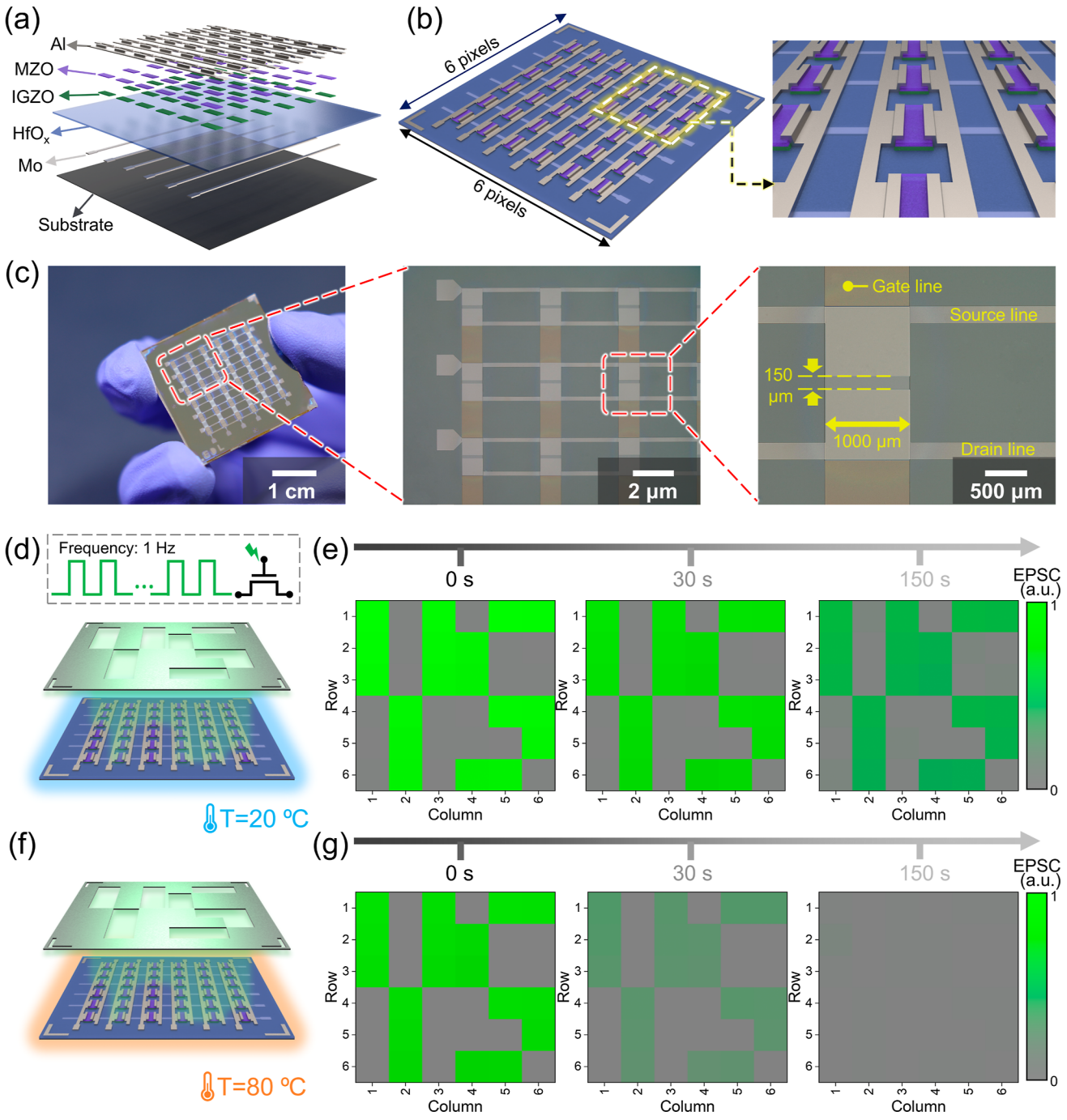



**Figure 5.** Array-level operation and temperature-dependent response. (a) Schematic illustration of the 6 × 6 TT-PST array structure. (b) Layout and enlarged view of the array configuration. (c) Optical and microscopic images of the fabricated array at different magnifications. (d) Schematic illustration of array operation under pulsed green-light stimulation ($\lambda$ = 532 nm, 1 Hz) at 20 °C. (e) Time-dependent spatial output maps at 20 °C, showing the evolution of the optically written "YS" pattern at 0, 30, and 150 s. (f) Schematic illustration of array operation under pulsed green-light stimulation at 80 °C. (g) Time-dependent spatial output maps at 80 °C, showing the evolution of the optically written "YS" pattern at 0, 30, and 150 s.

dielectric layer, an IGZO channel, an MZO modulation layer, and an Al top electrode, while maintaining the same MZO/IGZO heterostructure employed in the single-device platform. To verify that this transient synaptic behavior is maintained at the pixel-array level, double-pulse EPSC measurements were performed on three representative pixels randomly selected from the 6 × 6 array at 20, 50, and 80 °C (Figure S6). The selected pixels consistently exhibited the same temperature-dependent trend as the single-device platform, with more pronounced temporal accumulation and slower relaxation at lower temperature and faster decay after optical stimulation at higher temperature.

The overall array configuration and a zoomed-in view are shown in Figure 5b, indicating the scalability of the device into a regularly arranged multipixel platform. Optical and microscopic images at different magnifications (Figure 5c) reveal uniform pixel distribution and well-defined gate, source, and drain lines, with a channel length of approximately 150 $\mu$m and a width of 1000 $\mu$m. The electrical uniformity of the array is further confirmed by the transfer characteristics of all 36 pixels


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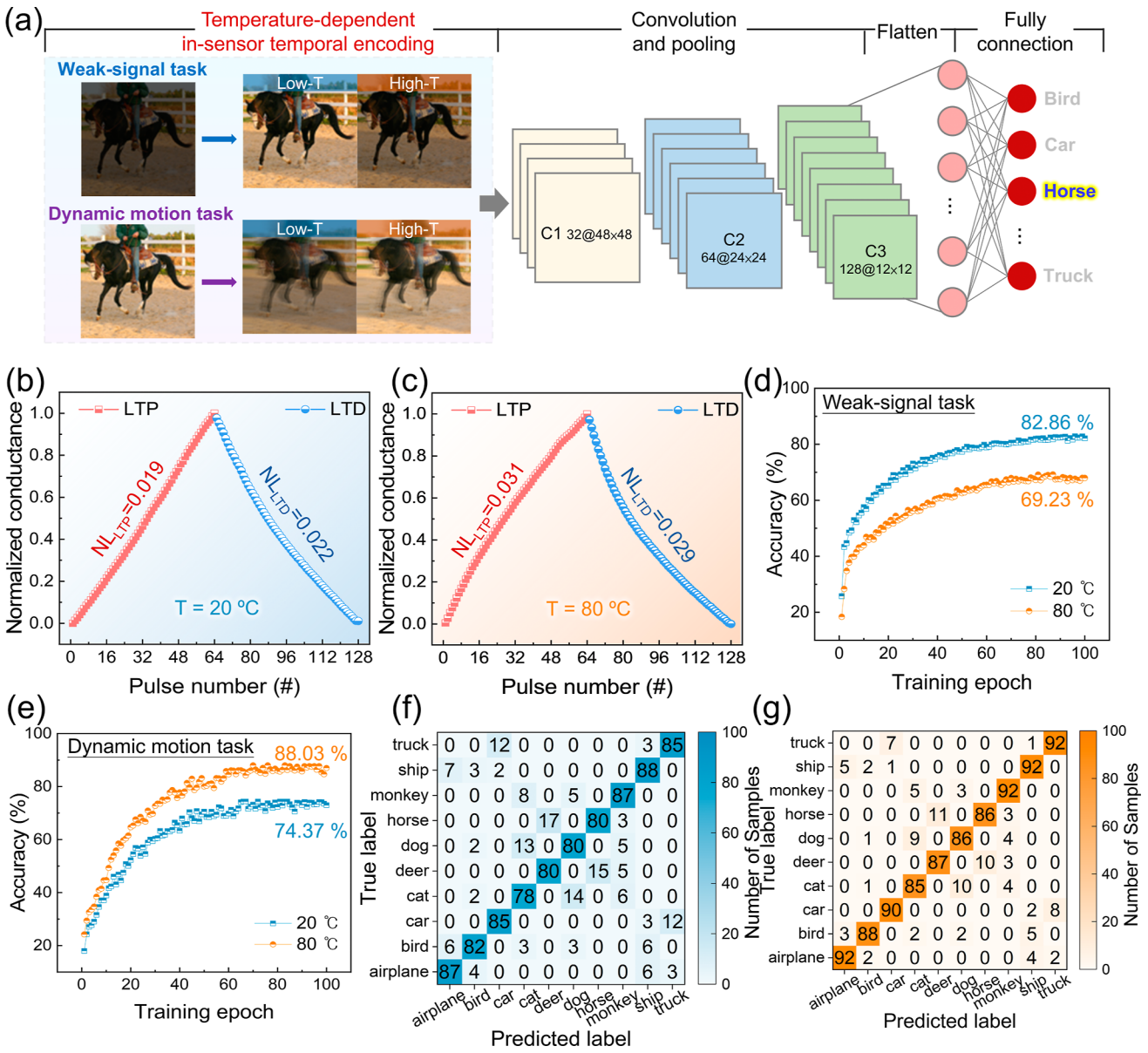



**Figure 6.** Temperature-adaptive temporal encoding and recognition. (a) Schematic illustration of the recognition framework integrating temperature-dependent in-sensor temporal encoding with a CNN, including weak-signal and dynamic motion tasks. (b,c) Measured LTP and LTD characteristics at 20 and 80 °C, respectively. (d) Recognition accuracy as a function of training epoch for the weak-signal task at different temperatures. (e) Recognition accuracy as a function of training epoch for the dynamic motion task at different temperatures. (f) Confusion matrix for the weak-signal task at 20 °C. (g) Confusion matrix for the dynamic motion task at 80 °C.

measured under identical conditions at $V_{DS}$ = 10.1 V (Figure S7), where both the threshold voltage and on-state current exhibit minimal variation, indicating excellent device-to-device consistency. The array operates under pulsed green-light stimulation (532 nm, 1 Hz) for 30 s, as illustrated in Figure 5d. The time origin ($t$ = 0 s) is defined as the moment immediately after the termination of the optical input. This operation scheme enables simultaneous encoding of spatial and temporal information. As a result, the array output not only reflects the spatial distribution of the input pattern but also preserves its temporal evolution, allowing the same hardware to exhibit distinct response behaviors across different time scales. At low temperature (20 °C), the array exhibits pronounced temporal integration and memory retention behavior (Figure 5e), with $V_G$ = − 2 V and $V_{DS}$ = 1.5 V. The input pattern (a predefined "YS" shape written by spatially selective optical stimulation) is initially encoded at $t$ = 0 s, and although the overall contrast gradually decreases at 30 s, the spatial features remain clearly distinguishable. Even after 150 s, the primary structure of the input pattern can still be identified. This behavior indicates a long effective retention time at low temperature, enabling persistent representation of historical inputs and resulting in an integration-dominant operating mode. At an elevated temperature of 80 °C, the array response undergoes a clear transition (Figure 5f,g), under the same bias conditions. The input pattern is rapidly established at the initial stage, but its contrast significantly diminishes at 30 s and nearly vanishes by 150 s. Under this condition, the array output primarily tracks the current input state, with minimal retention of past information. These results demonstrate that the same TT-PST array can operate in two distinct computational modes without any structural reconfiguration: an integration-dominant


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mode characterized by temporal integration and information retention at low temperature and a response-dominant mode featuring rapid response and dynamic updating at high temperature. This transition originates from the modulation of defect-mediated kinetic time scales, enabling a continuous shift between memory-enhanced and fast-tracking functionalities.

### 2.5. Temperature-Adaptive Temporal Encoding for Task-Oriented Recognition

To evaluate the task-level recognition capability of the proposed temperature-adaptive synaptic device, hardware-aware neural network simulations were performed using experimentally extracted device parameters. The measured decay dynamics was incorporated into the temporal input encoding stage, while the long-term potentiation (LTP)/long-term depression (LTD) characteristics were used to define synaptic weight updates during training. The STL-10 data set, containing 10 object categories (airplane, bird, car, cat, deer, dog, horse, monkey, ship, and truck), was used for all recognition tasks.[37] All images were used in their original 96 × 96 RGB format. Two representative tasks were designed to examine the functional impact of temperature-dependent temporal encoding. For the weak-signal task, image intensity was linearly attenuated to 20% of its original value to simulate low-contrast input, and the attenuated frame was repeatedly presented over multiple time steps to mimic continuous weak stimulation. For the dynamic motion task, each image was converted into a six-frame sequence using a fixed horizontal translation of 4 pixels per frame, with zero-padding applied to preserve spatial dimensions. These two tasks probe weak-signal accumulation and motion-sensitive processing, respectively. The temporal response of the device is governed by temperature-dependent carrier dynamics, which can be described as

$$S_t = \alpha S_{t-1} + (1 - \alpha)X_t \tag{2}$$

where $S_t$ is the accumulated response, $X_t$ is the input stimulus, and $\alpha$ is the temporal retention factor. The retention factor $\alpha$ is determined by the relaxation time constant $\tau$ as

$$\alpha = e^{-\Delta t/\tau} \tag{3}$$

where $\tau$ is the temperature-dependent relaxation time constant and $\Delta t$ is the time interval between successive inputs. The value of $\tau$, and thus $\alpha$, is extracted from experimentally measured decay characteristics.[38,39] As a result, the same input sequence exhibits different temporal responses at different temperatures. At low temperature (20 °C), a larger $\alpha$ leads to stronger temporal integration, facilitating the accumulation of weak signals while introducing residual contributions from previous frames. At high temperature (80 °C), a smaller $\alpha$ results in faster signal decay, reducing interframe interference but limiting signal accumulation. This transition should be understood as continuous rather than binary. Intermediate temperatures correspond to intermediate temporal-retention regimes, and the degree of temporal accumulation depends on the relationship between the temperature-dependent relaxation dynamics and the input interval $\Delta t$. Accordingly, the effective operating condition for weak-signal integration is jointly determined by temperature and signal timing conditions, rather than by a fixed temperature threshold. These differences lead to task-dependent encoding behavior, as illustrated by representative examples in Figure S8 (weak-signal task) and Figure S9 (dynamic motion task). LTP and LTD characteristics were experimentally measured at 20 and 80 °C, as shown in Figure 6b,c, respectively. LTP was induced by periodic green-light stimulation ($\lambda$ = 532 nm) at 1 Hz. LTD was implemented using positive gate voltage pulses at 1 Hz, with amplitudes of +1.5 V and +0.5 V at 20 and 80 °C, respectively. Different LTD pulse amplitudes were selected to obtain stable conductance-update characteristics for hardware-aware weight updating based on experimentally extracted device characteristics. For both LTP and LTD, 64 consecutive pulses were applied to obtain the full conductance evolution. The conductance evolution during LTP and LTD was fitted using exponential functions:

$$G_{LTP}(P) = B(1 - e^{-P/A}) + G_{min} \tag{4}$$

$$G_{LTD}(P) = -B(1 - e^{-(P_{max}-P)/A}) + G_{max} \tag{5}$$

$$B = \frac{G_{max} - G_{min}}{1 - e^{-P_{max}/A}} \tag{6}$$

where $G_{LTP}(P)$ and $G_{LTD}(P)$ denote the conductance at the $P$-th pulse during potentiation and depression, respectively. $P$ is the pulse number and $P_{max}$ is the total number of applied pulses (64 in this work). $G_{min}$ and $G_{max}$ represent the minimum and maximum conductance values, respectively. $A$ is the fitting parameter governing the approach to saturation, and $B$ is a scaling factor ensuring that the fitted curves span the full conductance range from $G_{min}$ to $G_{max}$.[40,41] From these fittings, key synaptic parameters, including dynamic range, number of conductance states, and nonlinearity (NL), were extracted and used as inputs to the neural network simulation (see Note S1 for definitions). The NL was quantified from the deviation of the conductance-update behavior from an ideal linear conductance update between $G_{min}$ and $G_{max}$. The extracted device parameters were incorporated into a hardware-aware neural network training framework to emulate hardware-constrained weight updating, while the task-selective behavior mainly originated from the temperature-dependent temporal dynamics of the TT-PST. In this implementation, each synaptic weight was mapped to a finite set of discrete conductance states defined by the measured $G_{min}$ and $G_{max}$, and weight updates followed the fitted LTP/LTD curves rather than ideal linear increments. This allows the training process to reflect the nonlinear and finite-resolution characteristics of the device. Considering the device-to-device variation and read/write operation errors in practical hardware systems, experimentally extracted device-to-device conductance dispersion from the fabricated array (with a maximum current variation of 6.7% across the 36-pixel array measurements) was additionally incorporated into the weight-update process through stochastic perturbation of the conductance states. The encoded representations were fed into a convolutional neural network (CNN) with three convolutional layers followed by a fully connected layer (Figure 6a). This architecture enables hierarchical feature extraction from the temporally encoded inputs and performs final classification across 10 object categories. For the weak-signal task, the network achieved higher recognition accuracy at 20 °C than at 80 °C, reaching 82.86% versus 69.23% (Figure 6d). This trend is consistent with stronger temporal integration at low temperature, which supports the accumulation of weak inputs. The corresponding confusion matrix is shown in Figure 6f. For the dynamic motion task, higher recognition accuracy was obtained at 80 °C (88.03%) compared with 20 °C (74.37%) (Figure 6e). This behavior is consistent with faster temporal decay at elevated temperature, which reduces interframe interference. The corresponding confusion matrix is shown in


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Figure 6g. Additional repeated-training runs showed only minor accuracy fluctuations while preserving the same temperature-dependent task-selective trends (Table S1). These results indicate that temperature can act as a practical tuning parameter for the temporal encoding behavior of the device. Through retention modulation, the same hardware platform shows distinct advantages in weak-signal and dynamic visual tasks, suggesting its promise for task-adaptive neuromorphic vision applications.

## 3. CONCLUSION

In this work, we present a thermally tunable photonic synaptic transistor (TT-PST) based on an MZO/IGZO heterostructure that harnesses temperature as a simple external stimulus to continuously and reversibly modulate synaptic temporal dynamics. Drawing inspiration from temperature-regulated temporal processing in biological systems such as hawkmoths, the device enables a smooth transition between integration-dominant and response-dominant modes through thermally accelerated detrapping of oxygen-vacancy states. As temperature increases from 20 to 80 °C, Max. PR rises from 0.43 to 2.25 A/W, accompanied by a pronounced reduction in temporal persistence. Array-level experiments and CNN-based simulations demonstrate that this single hardware platform can be optimally tuned for different visual tasks: low-temperature operation enhances weak-signal recognition in low-light conditions (82.86% accuracy), whereas high-temperature operation improves dynamic motion analysis (88.03% accuracy). These results highlight temperature as an effective, bioinspired control parameter for on-demand temporal engineering in photonic neuromorphic devices. The proposed TT-PST provides a practical route toward energy-efficient, task-adaptive neuromorphic vision systems without hardware reconfiguration, offering new opportunities for intelligent in-sensor computing in real-world environments.

## 4. EXPERIMENTAL METHODS

### 4.1. Device Fabrication

The TT-PST was fabricated on $p^+$-Si substrates with a thermally grown $SiO_2$ layer (about 120 nm), serving as the common gate and gate dielectric. An amorphous IGZO thin film with a thickness of approximately 40 nm was deposited by radio frequency (RF) magnetron sputtering using a ceramic target composed of $In_2O_3$, $Ga_2O_3$, and ZnO in a 1:1:1 molar ratio. The active channel region was defined using a shadow mask. The IGZO film was subsequently annealed at 400 °C for 1 h in ambient atmosphere. The MZO functional layer was prepared via a solution process. A precursor solution with a Mg/(Mg + Zn) ratio of 10 at % was prepared by dissolving zinc nitrate hexahydrate ($Zn(NO_3)_2{\cdot}6H_2O$, 0.2677 g) and magnesium nitrate hexahydrate ($Mg(NO_3)_2{\cdot}6H_2O$, 0.0256 g) in 2-methoxyethanol (2-ME, 5 mL). The solution was stirred until fully dissolved and then spin-coated onto the IGZO channel at 3000 rpm. The coated film was annealed at 300 °C for 1 h in an ambient atmosphere to form the MZO layer. Source and drain electrodes were deposited by RF magnetron sputtering of Al through a shadow mask, defining a channel width and length of approximately 1000 and 150 $\mu$m, respectively. For array-level devices, standard photolithography was employed to define the patterned electrodes and pixel layout.

### 4.2. Device Characterization

The structural, optical, and electronic properties of the films were characterized using multiple techniques. Optical microscope images were obtained using a digital microscope (VHX-7000, Keyence), and the cross-sectional morphology was examined by field-emission scanning electron microscopy (FE-SEM, JSM-7800F). The film thickness was measured by spectroscopic ellipsometry (alpha-SE, J.A. Woollam). Photoluminescence (PL) spectra were measured using a spectrofluorometer (Fluoromax Plus-C, Horiba) to evaluate the defect-related emission characteristics of MgZnO films with different Mg compositions. Ultraviolet−visible (UV−vis) absorption spectra were measured using a spectrophotometer (V-650, JASCO), and the optical bandgaps were extracted from the corresponding Tauc plots. Ultraviolet photoelectron spectroscopy (UPS, AXIS Supra, Kratos) was used to determine the electronic energy levels. The crystalline structure of the MZO films was analyzed by X-ray diffraction (XRD, SmartLab, Rigaku). The chemical states and composition were characterized by X-ray photoelectron spectroscopy (XPS, PHI 5000 Versa Probe, ULVAC-PHI).

### 4.3. Electrical Measurements

The electrical properties of the devices were evaluated by using a semiconductor parameter analyzer (4200A, Keithley Instruments). Transfer characteristics were measured by sweeping the gate voltage at a constant drain bias. Optoelectronic measurements were performed by using a laser source (Cubecell Technology) with a wavelength of 532 nm. For synaptic measurements, pulsed optical stimuli were applied at programmable frequencies, and the transient drain current was recorded as the EPSC. Temperature-dependent measurements were carried out by using a hot chuck system (MST-1000B, MS TECH).

### 4.4. Neural Network Simulation

Neural network simulations were performed by using experimentally extracted device parameters. The temporal response was modeled based on measured decay characteristics, and the synaptic weight update behavior was defined using fitted LTP and LTD curves. The extracted parameters were incorporated into the NeuroSim framework to emulate hardware-constrained weight updates. A convolutional neural network consisting of convolutional layers and a fully connected layer was employed for classification tasks using the STL-10 data set.

## ■ ASSOCIATED CONTENT

### Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsami.6c07920.

Synaptic parameter extraction; fabrication-process schematic; energy-band analysis of IGZO and MZO; wavelength-dependent photoresponse measurements; temperature-dependent kinetic analysis and Arrhenius-like fitting; thermal-cycling reversibility evaluation; array-level transient EPSC characterization and transfer characteristics; representative temperature-dependent encoded images for weak-signal and dynamic motion tasks; repeated-training results for neural network simulations; and supplementary array- and task-level evaluation data (PDF)

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[⊥]S.L. and H.J.K. contributed equally to this work. S.L. contributed to writing—original draft, conceptualization, methodology, and investigation. H.J.K. was responsible for device fabrication, experiments, and investigation. S.H.P. contributed to visualization. C.W. contributed to material characterization and data analysis. J.H., H.J.S., K.M.K., Y.W.K., and J.B.A. involved in the discussion, revision, and analysis of the results. J.M.M. provided guidance. H.J.K. was responsible for project administration and overall supervision.

## Notes

The authors declare no competing financial interest.

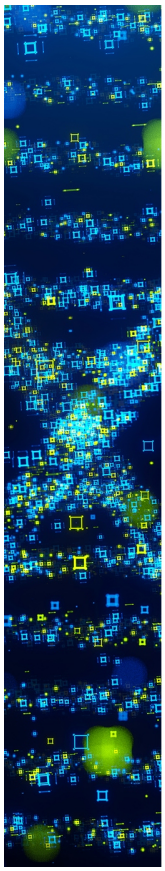
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